



State-of-Health (SOH) Estimation of Lithium-Ion Batteries Using Two-Stage Autoencoder and BPNN

Pipeline Pre-processing & Autoencoder Validation

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Dataset Overview

Battery ID	Total Cycles	Total Impedance Records	Initial Capacity (Ah)	Final Capacity (Ah)
B0005 (1)	614	10,842	1.85	1.32
B0006 (2)	614	10,842	2.03	1.18
B0007 (3)	614	10,842	1.89	1.43
B0018 (4)	319	2,067	1.85	1.34

- ❑ **Data Source:** We are using 4 lithium-ion batteries from the NASA dataset.
- ❑ **Data Types:** The dataset records Electrochemical Impedance Spectroscopy (EIS) and Capacity (Ah) for each cycle.
- ❑ **Objective:** Our AI pipeline uses the Impedance records to predict this exact capacity drop (State of Health - SOH).

Battery_ID	Cycle	Capacity_Ah
1	2	1.856487
1	4	1.846327
1	6	1.835349
1	8	1.835263
1	10	1.834646
1	12	1.835662
1	14	1.835146
1	16	1.825757
1	18	1.824774
1	20	1.824613
1	22	1.82462

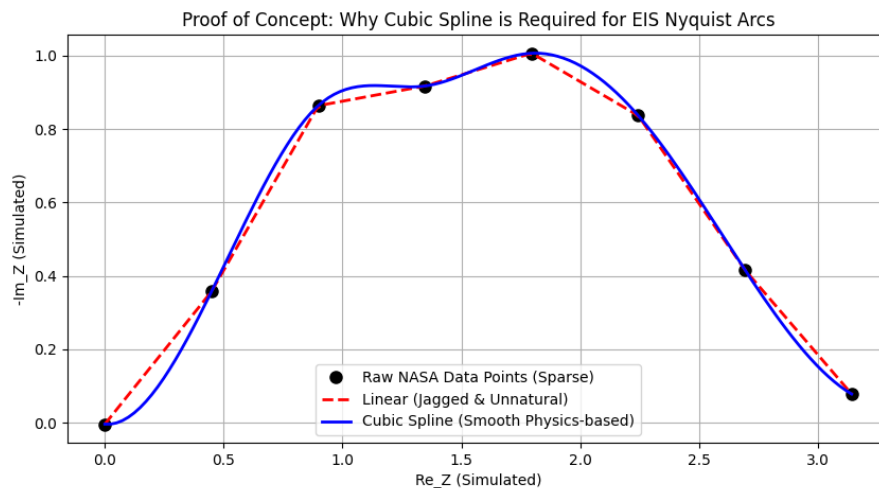
Battery_ID	Cycle	Re_Z	Neg_Im_Z
1	41	0.070069	0.00048
1	41	0.068179	0.00119
1	41	0.067933	5.68E-05
1	41	0.066918	0.000879
1	41	0.068071	0.000197
1	41	0.066571	0.000584
1	41	0.068418	0.000957
1	41	0.067724	0.000477
1	41	0.065793	0.000645
1	41	0.066825	0.000803
1	41	0.066266	0.000883
1	41	0.067426	0.00123
1	41	0.067177	0.001568
1	41	0.065518	0.00114
1	41	0.066375	0.000996
1	41	0.066364	0.001631
1	41	0.063881	0.000731



Spatial Pre-processing (Data Interpolation)

Step 1: Physics-Based Data Interpolation

- **Goal:** Resize raw data to 256 features (16x16) for the Autoencoder.
- **Problem:** Linear interpolation creates artificial jagged edges (noise).
- **Solution:** Used **Cubic Spline** interpolation.
- **Result:** Preserves the true physical shape of the electrochemical Nyquist arc.

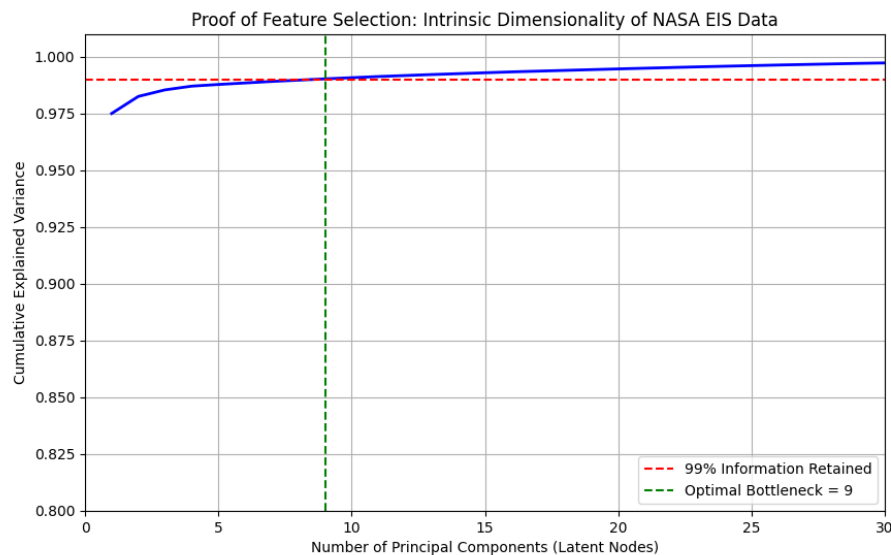




Feature Selection (Intrinsic Dimensionality)

Step 2: Finding the Optimal Bottleneck Size

- ❑ **Goal:** Find the exact number of latent nodes needed for the Autoencoder.
- ❑ **Method:** Principal Component Analysis (PCA).
- ❑ **Result:** 9 components retain 99% of the original variance.
- ❑ **Conclusion:** Bottleneck size is mathematically set to 9. Using 20 nodes causes redundancy.

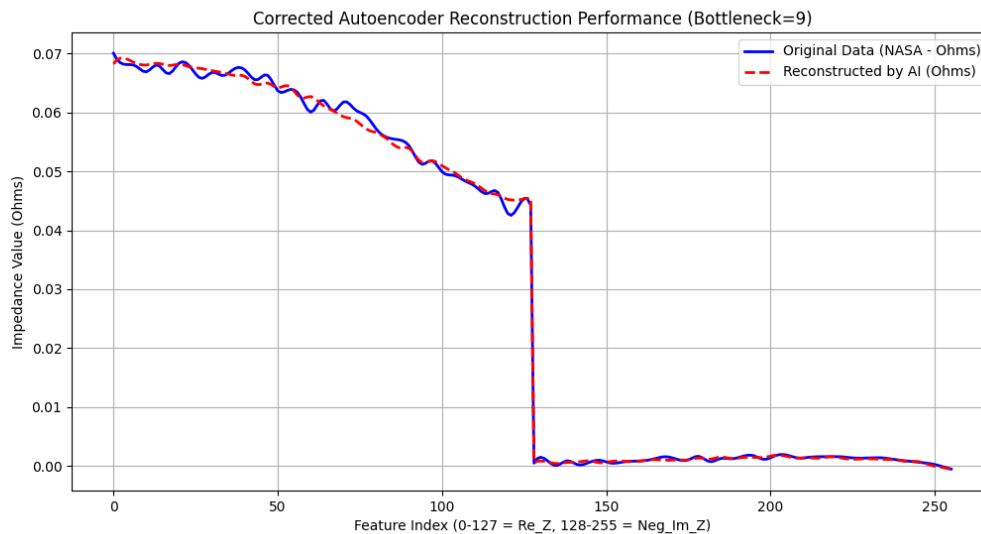




Autoencoder Spatial Performance

Step 3: Autoencoder Reconstruction Results

- ❑ **Model:** Standard Autoencoder (9 Latent Nodes).
- ❑ **Result:** The reconstructed signal (red) perfectly matches the original signal (blue) within 1 cycle.
- ❑ **Conclusion:** The AE compresses and rebuilds the spatial features without data loss.

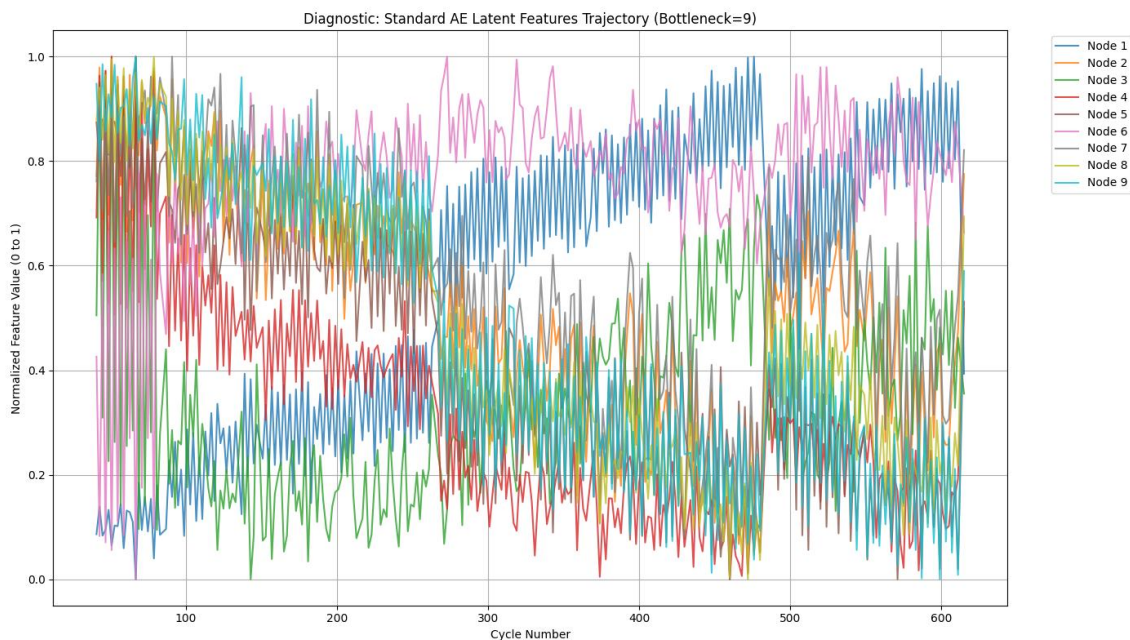




Diagnostic of Latent Space

Step 4: The Temporal Jitter Issue

- ❑ **Observation:** The 9 latent nodes show high-frequency "sawtooth" patterns over time (cycles).
- ❑ **Analysis:** The Autoencoder is a *stateless* spatial model. It processes one cycle at a time.
- ❑ **Hypothesis:** The jitter is not a model error. It comes directly from the raw data.

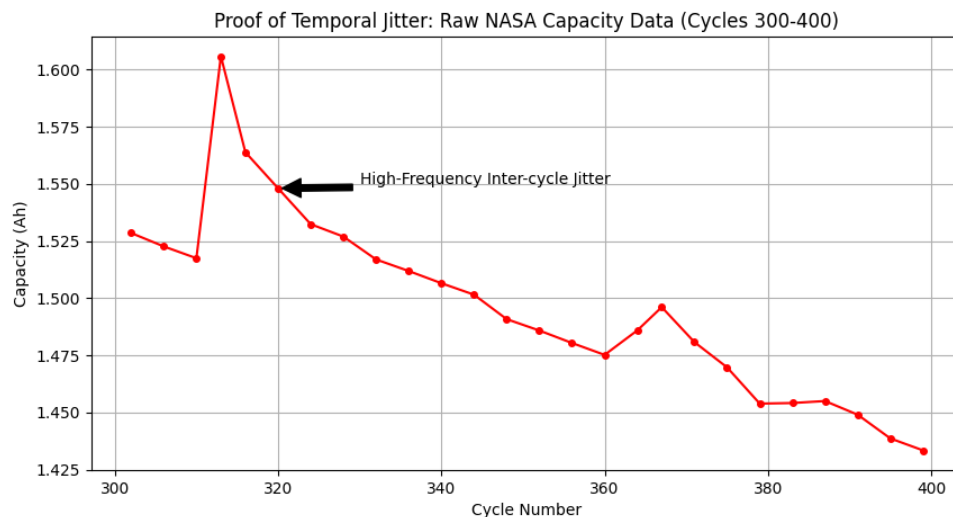




Root Cause Analysis (Proving the Hypothesis)

Root Cause: Inherent Dataset Noise

- ❑ **Evidence:** Zooming in on the raw NASA Capacity data (Cycles 300-400).
- ❑ **Fact:** The raw data inherently jumps up and down between cycles (Rest time recovery, sensor error).
- ❑ **Conclusion:** We cannot use "Moving Average" to fix this, as it causes **Data Leakage** (using future data).





Next Steps (Moving to BPNN)

Next Steps: Handling Temporal Noise in BPNN

- ❑ **Current Status:** AE Pipeline (Stage 1) is fully verified and locked (Bottleneck = 9).
- ❑ **Next Phase:** Train the BPNN (Regression model) to find the global SOH trend.
- ❑ **Strategy:**
 - Use a **Shallow Network** to avoid memorizing the sawtooth noise.
 - Apply **L2 Regularization (Weight Decay)** to force the model to find the smooth monotonic degradation trend.